

IRAKO, Japan

WMO: 476530

Lat: 34.6297N

Lon: 137.0939E

Elev: 25

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.4	34.9	17.0	13.0	38.9	19.4	14.6	39.6	27.3	43.5	24.7	41.7	10.6	340	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	91.4	78.5	89.1	78.1	87.1	77.6	80.8	86.7	79.9	85.5	79.0	84.5	9.2	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.3	152.3	83.9	78.3	147.1	83.1	77.4	142.5	82.4	44.5	86.8	43.5	85.7	42.6	84.7	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.9	20.3	18.3	DB	29.7	95.4	1.6	2.3	28.6	97.0	27.7	98.4	26.8	99.6	25.7	101.3
			WB	26.3	81.8	1.7	1.5	25.1	82.9	24.1	83.8	23.1	84.7	21.9	85.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	42.8	43.6	49.0	58.0	66.3	72.4	79.9	82.5	76.6	67.0	56.5	47.4	(6)
	DBStd	14.29	3.53	4.19	4.74	4.95	3.52	3.71	3.87	3.11	4.31	4.52	4.82	4.48	(7)
	HDD50	605	226	186	78	4	0	0	0	0	0	0	4	107	(8)
	HDD65	2861	689	600	496	215	27	1	0	0	0	30	256	547	(9)
	CDD50	4957	2	6	47	243	505	673	926	1007	797	527	199	25	(10)
	CDD65	1737	0	0	0	4	67	223	461	542	347	92	1	0	(11)
	CDH74	13529	0	0	0	6	146	854	4101	5788	2430	204	0	0	(12)
	CDH80	3961	0	0	0	0	10	127	1262	2045	507	10	0	0	(13)
(14) Wind	WSAvg	8.5	10.5	10.1	9.7	8.9	8.1	7.0	6.9	7.2	7.5	7.6	8.3	9.8	(14)
(15) Precipitation	PrecAvg	64.1	2.4	3.0	4.6	5.5	6.6	7.3	6.2	5.0	10.0	7.0	4.3	2.4	(15)
	PrecMax	86.9	7.4	7.6	7.8	10.4	14.3	12.8	13.1	12.3	19.8	25.2	8.6	6.0	(16)
	PrecMin	40.9	0.0	0.9	1.9	1.0	2.8	2.6	0.4	0.6	2.6	1.2	0.4	0.2	(17)
	PrecStd	11.6	1.9	1.6	1.5	2.4	2.8	2.2	3.5	3.1	4.2	4.5	2.3	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	61.1	66.3	74.2	81.3	87.1	93.6	95.2	89.9	81.4	71.2	63.8	(19)
		MCWB	49.7	55.6	56.2	61.2	66.6	75.5	78.4	78.9	76.8	71.3	62.9	58.8	(20)
	2%	DB	52.9	56.5	63.1	71.1	77.5	83.4	90.9	92.1	86.9	78.3	68.6	59.2	(21)
		MCWB	46.1	49.4	55.6	61.2	65.2	73.4	78.2	78.7	76.0	69.7	61.4	53.1	(22)
	5%	DB	50.8	53.3	60.6	68.7	75.3	80.8	88.4	90.1	84.9	76.3	66.6	56.8	(23)
		MCWB	44.3	46.6	53.8	60.1	64.7	71.7	78.0	78.6	75.5	68.3	60.0	50.4	(24)
	10%	DB	48.9	50.8	57.9	66.5	73.3	78.5	86.3	88.2	82.9	74.2	64.6	54.7	(25)
		MCWB	42.6	44.4	51.0	58.8	63.8	70.6	77.5	78.2	74.8	66.6	58.3	48.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	57.9	61.0	68.0	72.1	78.0	81.8	82.5	80.0	75.6	66.7	60.2	(27)
		MCDB	54.6	60.1	62.6	69.3	75.4	83.0	88.6	89.1	85.3	77.8	68.3	62.6	(28)
	2%	WB	47.6	51.2	58.0	64.7	69.5	76.1	80.4	81.2	78.7	72.8	63.8	54.7	(29)
		MCDB	51.1	54.8	61.5	68.2	73.0	80.9	86.5	87.2	83.3	76.2	66.7	57.9	(30)
	5%	WB	45.4	48.0	55.0	62.5	68.0	74.4	79.3	80.4	77.7	70.4	61.6	51.8	(31)
		MCDB	49.5	51.8	59.2	66.6	72.2	78.9	85.0	86.1	82.2	74.5	65.2	55.9	(32)
	10%	WB	43.5	45.5	52.1	60.3	66.2	72.7	78.4	79.6	76.7	67.9	59.4	49.4	(33)
		MCDB	48.1	49.8	56.4	64.8	70.9	76.9	83.9	85.2	81.1	72.8	63.5	53.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.7	13.6	14.1	13.0	10.9	10.7	11.3	10.6	11.0	11.3	10.6	(35)
		MCDBR	13.2	15.5	17.9	17.1	16.5	13.8	13.9	13.7	12.2	12.2	13.1	13.3	(36)
	5% WB	MCWBR	9.4	11.0	12.3	9.8	8.2	6.2	5.1	4.8	5.4	7.2	9.1	9.8	(37)
		MCWBR	12.5	13.8	15.6	13.0	12.6	11.3	12.1	11.9	10.3	10.3	11.6	12.1	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.387	0.430	0.490	0.534	0.541	0.563	0.556	0.528	0.492	0.465	0.417	0.386	(40)
		taud	2.220	2.112	1.960	1.902	1.955	1.973	2.042	2.110	2.179	2.171	2.217	2.238	(41)
	Ebn at Noon	Ebn at Noon	252	253	247	243	243	237	238	242	245	239	239	244	(42)
		Edh at Noon	35	44	55	62	59	58	54	49	44	41	35	33	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	833	1038	1333	1550	1666	1482	1637	1681	1307	1032	841	747	(44)
	RadStd	RadStd	53	75	87	157	180	141	215	146	117	109	75	49	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page